Name _____ Period _____ Date _____

3-2 Lesson Quiz: Adding, Subtracting, and Multiplying Polynomials

short form questions keep their Savvas numbers (1, 2, 4); show your work

1. Add $(4a^2b - 3ab^2 + 2ab + 5)$ and $(2a^2b + 3ab^2 - 7ab)$. Complete the expression in standard form. Write one tile in each box.

a^2b	$6a^2b$	$6ab^2$	$(-9ab^2)$	$(-5ab)$	$9ab$	$5ab$	5

$$\square + \square + \square$$

Work:

2. Multiply $(x + 2y)(x^2 - xy + 3y)$.

Ⓐ $x^3 + x^2y + 3xy - 2xy^2 + 6y^2$

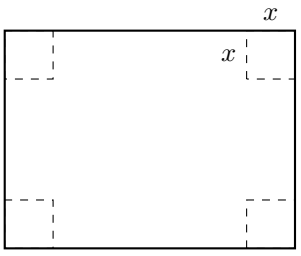
Ⓑ $x^3 - 3x^2y + 3xy - 2xy^2 + 6y^2$

Ⓒ $x^3 + 3xy - xy^2 + 3y^2$

Ⓓ $x^3 - xy + 6y^2$

Work:

4. Jae is constructing an open box from a piece of cardboard that is 9 in. wide and 12 in. long. Jae cuts squares of equal size from each corner of the cardboard, and then folds up the sides of the box. Write and simplify a polynomial function V in standard form for the volume of the box in terms of x.



$V(x) =$ _____ $x^3 -$ _____ $x^2 +$ _____ x

Work:

Scoring: Question 2 and Question 4 are 10 points each. Question 1 is a bonus of up to 5 points. Total out of 20.

3-2 Lesson Quiz, short form: answer key

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1. $6a^2b + (-5ab) + 5$**.** Like terms: $4a^2b + 2a^2b = 6a^2b$; $-3ab^2 + 3ab^2 = 0$ (the ab^2 terms cancel, so no ab^2 tile is used); $2ab - 7ab = -5ab$; the constant is 5. Standard form: degree 3, then degree 2, then the constant. Savvas quiz item 1, DOK 1, Skills Review and Practice A18 (IXL 9A3). Example 1 teaches it and is skipped in the fast packet; the 9A3 Jam and the subtraction step in Example 4 (distribute -1, combine like terms) carry it. **Bonus 5:** 1 per correct box in the correct order (3), 2 for shown grouping of like terms.

2. **A,** $x^3 + x^2y + 3xy - 2xy^2 + 6y^2$**.** $x(x^2 - xy + 3y) + 2y(x^2 - xy + 3y) = x^3 - x^2y + 3xy + 2x^2y - 2xy^2 + 6y^2$, and $-x^2y + 2x^2y = x^2y$. Choice B subtracts $2x^2y$ instead of adding it ($-3x^2y$); C and D leave out the x^2y term and do not show all six products. Savvas quiz item 2, DOK 1, A20 (IXL 8GN). Taught by Example 2A. **10 points:** 6 for A, 4 for work that shows all six products.

4. $V(x) = 4x^3 - 42x^2 + 108x$**.** Length $12 - 2x$, width $9 - 2x$, height x: $V(x) = x(12 - 2x)(9 - 2x) = x(108 - 24x - 18x + 4x^2) = x(4x^2 - 42x + 108)$. Savvas quiz item 4, DOK 3, A20 (IXL 8GN). Taught by Example 4 (write a polynomial function from a situation), Example 2B (three factors, two first) and the packet Practice (TE Practice 27, the foundry tray, filed under Example 4 in the TE item analysis). **10 points:** 3 for the three dimensions (1 each), 1 for the product $x(12-2x)(9-2x)$, 6 for the simplified standard form (2 per coefficient).

Items left off this form (available for a retake or Form B): Savvas item 3, closure of monomials under multiplication (answer C, "Yes; the product of any two monomials is a monomial"; DOK 2, A20; it is Try It 3 of the cut Example 3); Savvas item 5, hemisphere $V(x) = \frac{2}{3}\pi x^3$ versus the graphed cube volume over $[2, 3]$ (answer: the hemisphere is increasing faster; DOK 2, F61; average rate of change, the cut Example 5 and Lesson 3-1). Item 5's graph ($y = x^3$, window $0 \le x \le 5$, $0 \le y \le 100$) could be redrawn; it is left off because no kept Example teaches it, not because of the figure.

IXL replacement: SmartScore 70 or above on 8GN and 9A3 (the starred skills) replaces this quiz with 20/20.